

Orion Main

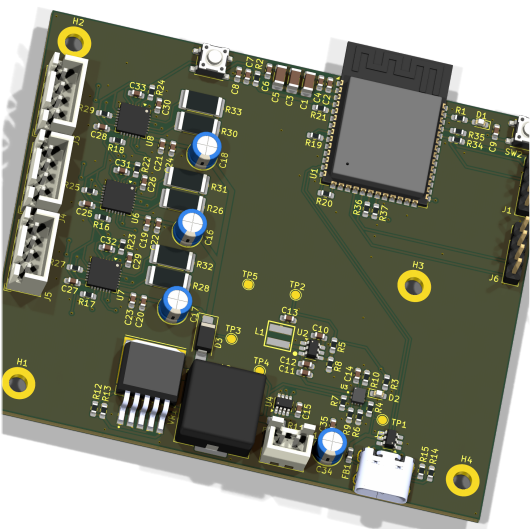
Variant:

2026-10-04

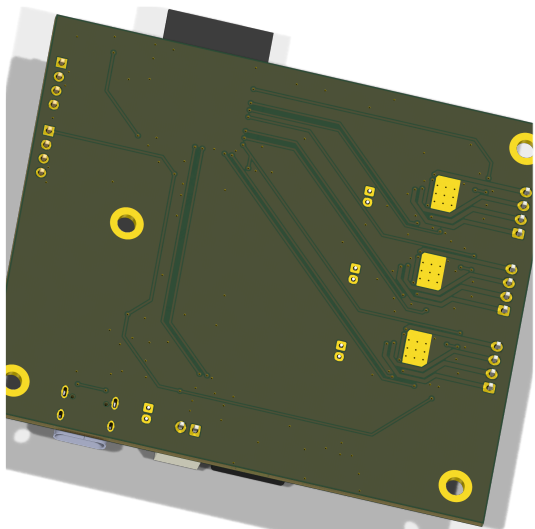
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as X

- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

Date: 04-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

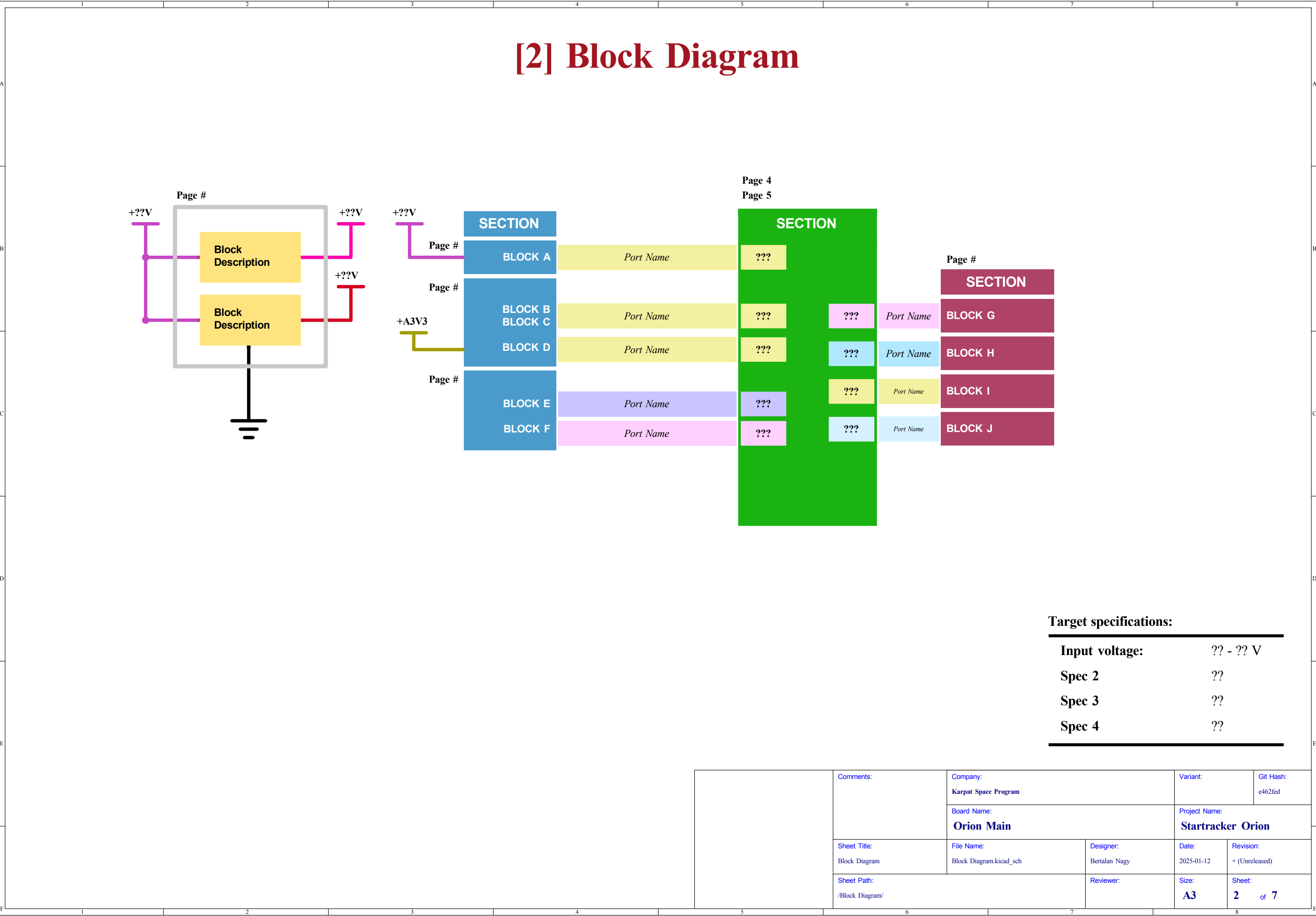
DESIGN NOTE:

Example text for critical design notes.

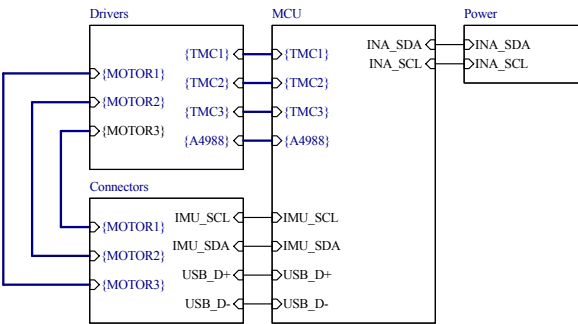
LAYOUT NOTE:

Example text for critical layout guidelines.

|  |                  |                                  |                            |                                    |                             |
|--|------------------|----------------------------------|----------------------------|------------------------------------|-----------------------------|
|  | Comments:        | Company:<br>Karpát Space Program |                            | Variant:                           | Git Hash:<br>e462fed        |
|  | Sheet Title:     | Board Name:<br>Orion Main        |                            | Project Name:<br>Startracker Orion |                             |
|  | Sheet Path:<br>/ | File Name:<br>Orion.kicad_sch    | Designer:<br>Bertalan Nagy | Date:<br>2025-01-12                | Revision:<br>+ (Unreleased) |
|  |                  |                                  | Reviewer:                  | Size:<br>A3                        | Sheet:<br>1 of 7            |

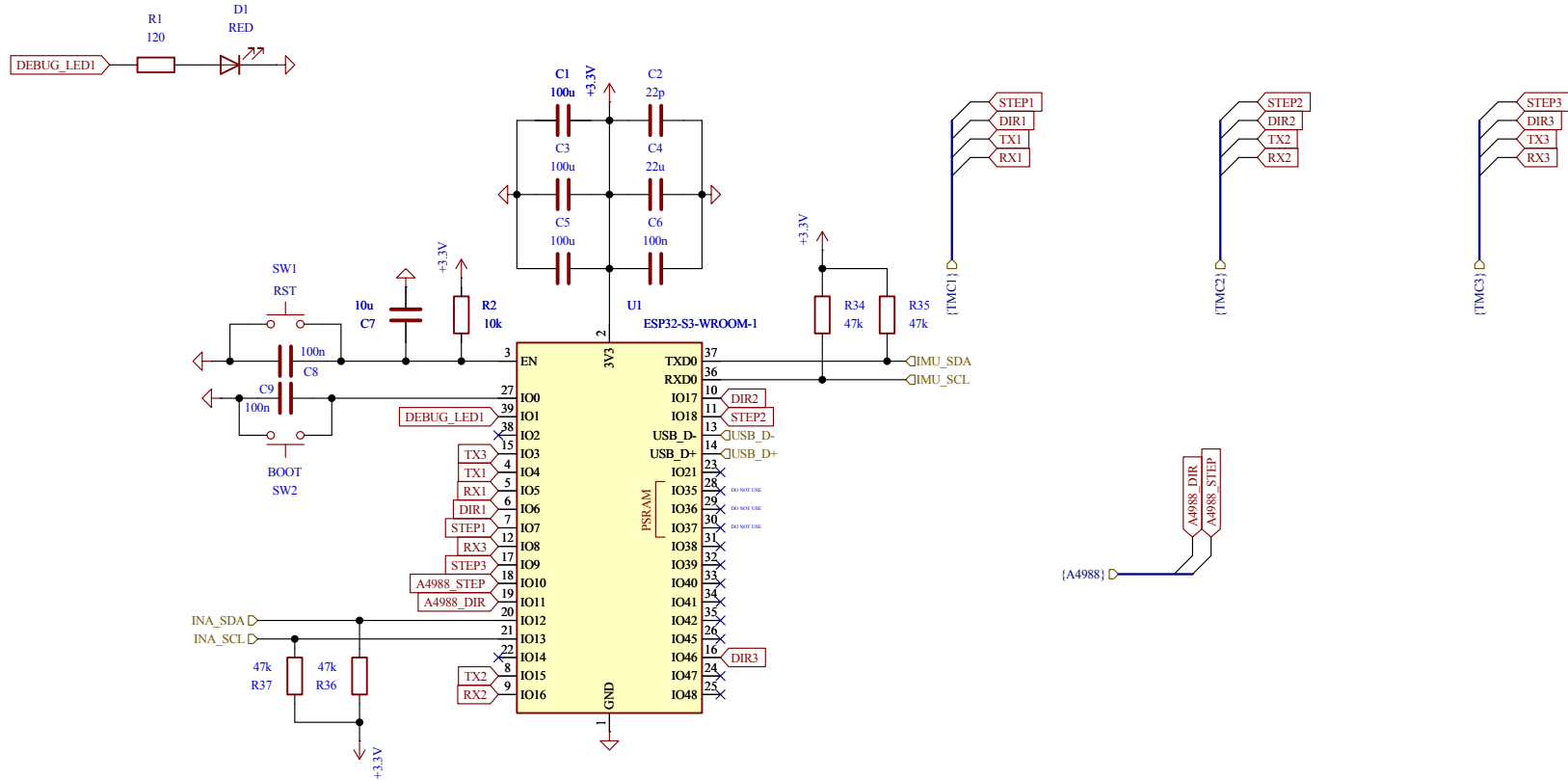


# [3] Project Architecture



|  |                                       |                                              |                            |                                    |                             |
|--|---------------------------------------|----------------------------------------------|----------------------------|------------------------------------|-----------------------------|
|  | Comments:                             | Company:<br>Karpát Space Program             |                            | Variant:                           | Git Hash:<br>e462fed        |
|  |                                       | Board Name:<br>Orion Main                    |                            | Project Name:<br>Startracker Orion |                             |
|  | Sheet Title:<br>Project Architecture  | File Name:<br>Project Architecture.kicad_sch | Designer:<br>Bertalan Nagy | Date:<br>2025-01-12                | Revision:<br>+ (Unreleased) |
|  | Sheet Path:<br>/Project Architecture/ |                                              | Reviewer:                  | Size:<br>A3                        | Sheet:<br>3 of 7            |

# [4] MCU



|  |                                           |                                  |                            |                                           |                                |
|--|-------------------------------------------|----------------------------------|----------------------------|-------------------------------------------|--------------------------------|
|  | Comments:                                 | Company:                         |                            | Variant:                                  | Git Hash:<br>e462fed           |
|  |                                           | Board Name:<br><b>Orion Main</b> |                            | Project Name:<br><b>Startracker Orion</b> |                                |
|  | Sheet Title:                              | File Name:<br>mcu.kicad_sch      | Designer:<br>Bertalan Nagy | Date:                                     | Revision:                      |
|  | Sheet Path:<br>/Project Architecture/MCU/ |                                  | Reviewer:                  | Size:<br><b>A4</b>                        | Sheet:<br><b>4</b> of <b>7</b> |

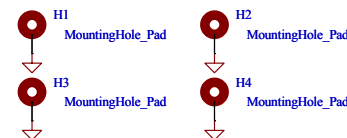
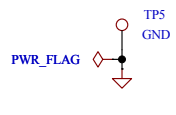
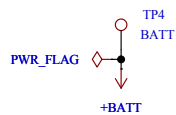
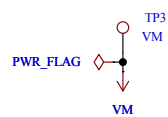
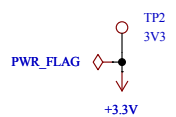
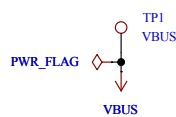
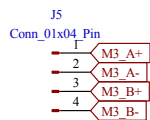
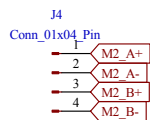
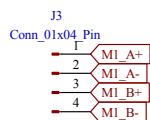
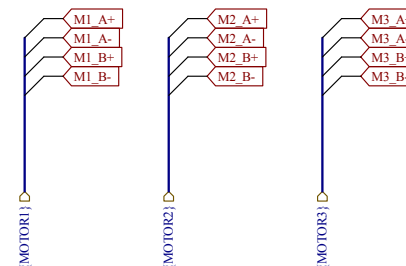
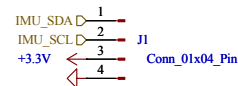
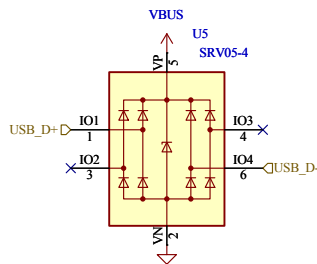
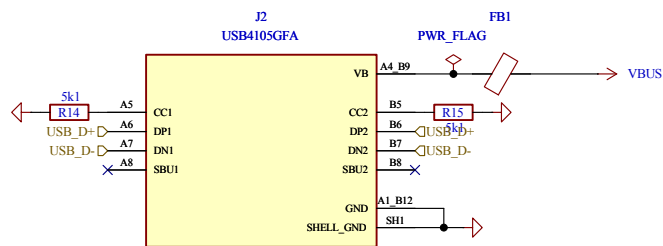
# [5] Power

The circuit schematic for the Orion Main board power section is shown. It includes a +BATT input, a +3.3V output, and a VM output. The circuit uses a TPS2121RUXR (U3) for current limiting and a LMR51420 (U2) for voltage regulation. A battery (BT1) is connected to the +BATT input. The circuit also includes a current sense resistor (R11) and an INA219BxDCN (U4) for monitoring. A final output VM is provided by a voltage divider (R12, R13) and a diode (D3).

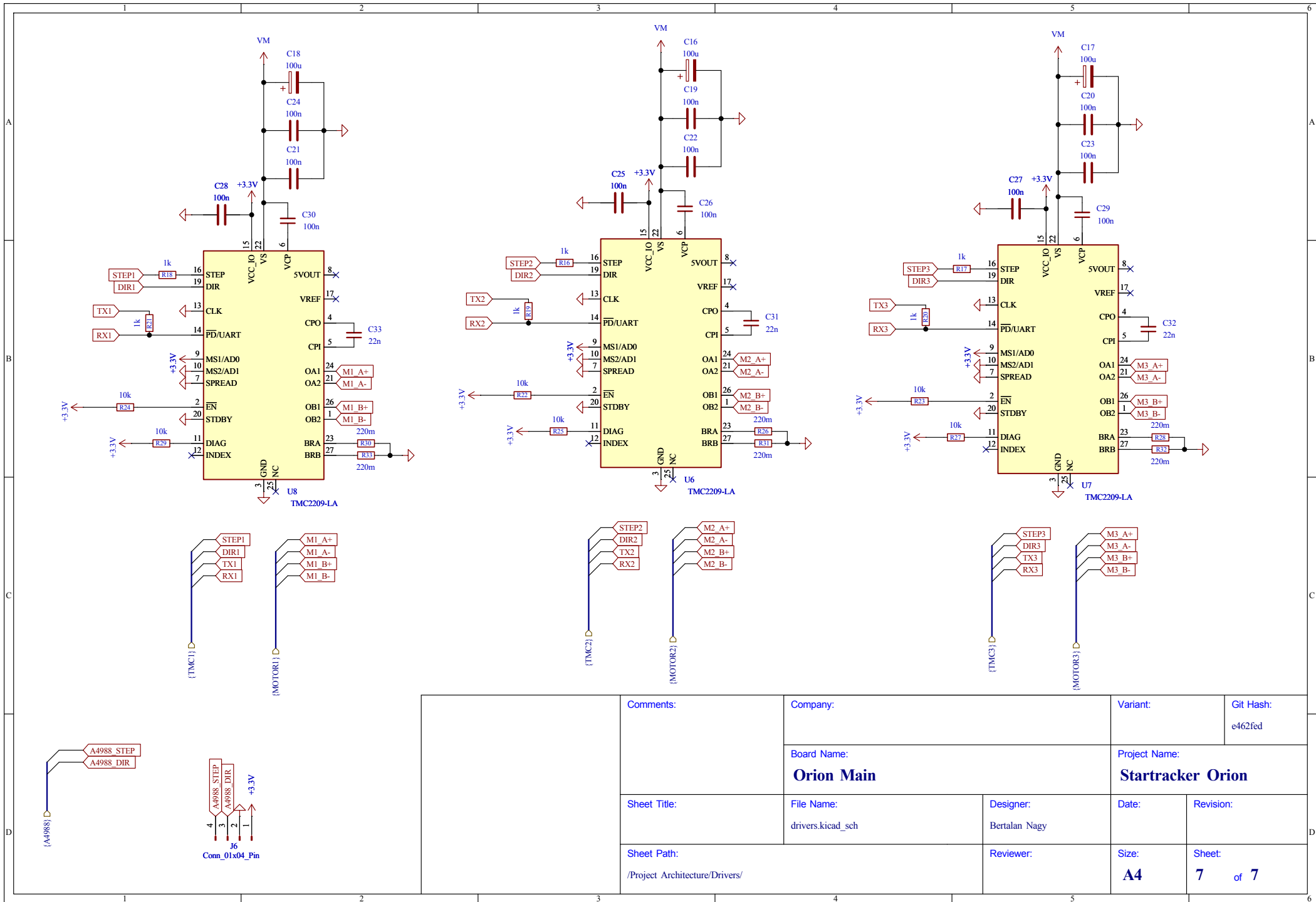
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| Sheet Title:                 | File Name:      | Designer:     | Date:     |
|                              | power.kicad_sch | Bertalan Nagy |           |
| Sheet Path:                  | Reviewer:       | Size:         | Sheet:    |
| /Project Architecture/Power/ |                 | A4            | 5 of 7    |

|                                             |                                  |                            |                                           |                                |
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|                                             | Board Name:<br><b>Orion Main</b> |                            | Project Name:<br><b>Startracker Orion</b> |                                |
| Sheet Title:                                | File Name:<br>power.kicad_sch    | Designer:<br>Bertalan Nagy | Date:                                     | Revision:                      |
| Sheet Path:<br>/Project Architecture/Power/ |                                  | Reviewer:                  | Size:<br><b>A4</b>                        | Sheet:<br><b>5</b> of <b>7</b> |

# [6] Connectors



|                                                  |                                  |                            |                                           |                                |
|--------------------------------------------------|----------------------------------|----------------------------|-------------------------------------------|--------------------------------|
| Comments:                                        | Company:                         |                            | Variant:                                  | Git Hash:<br>e462fed           |
|                                                  | Board Name:<br><b>Orion Main</b> |                            | Project Name:<br><b>Startracker Orion</b> |                                |
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Comments:

Company:

Variant:

Git Hash:

e462fed

Board Name:

Orion Main

Project Name:

Startracker Orion

Sheet Title:

File Name:

drivers.kicad\_sch

Designer:

Bertalan Nagy

Date:

Revision:

Sheet Path:

/Project Architecture/Drivers/

Reviewer:

Size:

A4

Sheet:

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